Shiv Nadar University Chennai
CS3807 Deep Learning Laboratory

Experiment 4 : Comparative Study of Deep Convolutional Neural Network Architectures Using Transfer Learning

Degree & Branch	B.Tech Artificial Intelligence & Data Science	Semester V
Subject Code & Name,Details	CS3807 – Deep Learning Laboratory Kalpana Bhaskar (AI-DS:A, Roll no: 24011101049)	AY: 2026–27

1 Objective

The objectives of this experiment are

- Study the evolution of deep CNN architectures.
- Compare LeNet-5, AlexNet, VGG16, GoogleNet and ResNet.
- Understand transfer learning.
- Fine tune pretrained CNN models.
- Compare classification performance of different architectures.

2 Learning Outcomes

After completing this experiment students will be able to

1. Explain modern CNN architectures.
2. Differentiate LeNet, AlexNet, VGG16, GoogleNet and ResNet.
3. Implement transfer learning.
4. Fine tune pretrained CNN models.
5. Compare CNN architectures using performance metrics.

3 Introduction

Deep Convolutional Neural Networks have become the dominant models for computer vision applications because they automatically learn hierarchical features from images. The evolution of CNNs has focused on improving classification accuracy while reducing computational complexity and training difficulties.

The progression of CNN architectures began with LeNet-5 and has evolved through AlexNet, VGGNet, GoogleNet, and ResNet. Each architecture introduced important innovations that significantly improved image classification performance.

4 Evolution of CNN Architectures

Model	**Year**	**Major Contribution**
LeNet-5	1998	First practical convolutional neural network
AlexNet	2012	ReLU activation, Dropout and GPU training
VGG16	2014	Uniform 3×3 convolution filters
GoogleNet	2014	Inception modules for multi-scale feature extraction
ResNet	2015	Residual learning using skip connections

5 LeNet-5

LeNet-5 was proposed by Yann LeCun in 1998 for handwritten digit recognition. It consists of two convolution layers, two average pooling layers and fully connected layers. It demonstrated that convolutional networks could automatically learn image features without handcrafted feature extraction.
Advantages

- Small number of parameters
- Fast training
- Suitable for OCR

6 AlexNet

AlexNet won the ImageNet Challenge in 2012 by reducing the classification error significantly. It introduced ReLU activation, Dropout regularization and GPU training.
Major innovations

- ReLU activation
- Dropout
- GPU implementation
- Data augmentation

7 VGG16

VGG16 demonstrated that using multiple small 3×3 convolution filters could outperform larger filters while increasing network depth.
Advantages

- Excellent feature extraction
- Simple architecture
- High classification accuracy

8 Comparison of Architectures

Architecture	**Depth**	**Parameters**	**Main Contribution**
LeNet-5	7	60K	First CNN
AlexNet	8	61M	ReLU + Dropout
VGG16	16	138M	Deep 3×3 filters
GoogleNet	22	6.8M	Inception Modules
ResNet50	50	25.6M	Residual Learning

9 GoogleNet (Inception Network)

GoogleNet, proposed by Szegedy et al. in 2014, introduced the Inception Module, which performs multiple convolution operations in parallel using different filter sizes. Instead of using only a single kernel size, GoogleNet simultaneously applies 1×1, 3×3, 5×5 convolutions and max pooling. The outputs are concatenated to obtain multi-scale feature representations.

The use of 1×1 convolutions before larger kernels reduces the number of trainable parameters and computational complexity.
Advantages

- Multi-scale feature extraction.
- Fewer parameters than VGG16.
- High computational efficiency.
- Better classification performance.

10 ResNet

Increasing CNN depth often causes the **vanishing gradient problem**, making optimization difficult. ResNet solves this issue using **skip (residual) connections**, allowing gradients to flow directly through the network.

Instead of learning

$$H(x)$$

ResNet learns

$$F(x) = H(x) - x$$

thus,

$$Output = F(x) + x$$

where

- $F(x)$ = Residual Mapping
- x = Identity Mapping

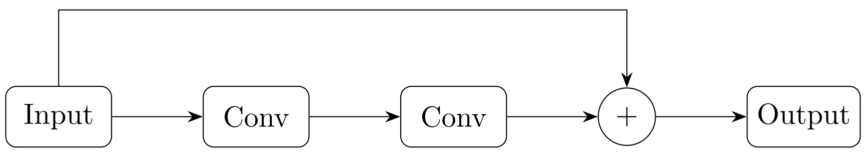

Figure 1: Residual Block in ResNet

Advantages

- Eliminates vanishing gradients.
- Enables very deep neural networks.
- Faster convergence.
- High ImageNet accuracy.

11 Dilated Convolution

Dilated convolution (also called atrous convolution) increases the receptive field without increasing the number of parameters. The dilation rate determines the spacing between kernel elements.
For a standard convolution, $D = 1$
For dilated convolution, $D > 1$
where zeros indicate inserted gaps.
Example

$$\begin{bmatrix} 1 & 0 & 2 \\ 0 & 3 & 0 \\ 4 & 0 & 5 \end{bmatrix}$$

Applications

- Semantic segmentation
- Medical image analysis
- Satellite imagery
- Object localization

12 Transpose Convolution

Transpose convolution increases the spatial dimensions of feature maps. Unlike pooling, which reduces image size, transpose convolution performs learnable upsampling.
Applications

- Image Super Resolution
- Autoencoders
- GANs
- Image Generation
- Semantic Segmentation

13 Transfer Learning

Transfer Learning utilizes a pretrained CNN model that has already learned general image features from a large dataset such as ImageNet.

Instead of training from scratch,

1. Load a pretrained model.
2. Remove the original classifier.
3. Freeze convolution layers.
4. Add new Dense layers.
5. Train the classifier.
6. Fine tune selected convolution layers.

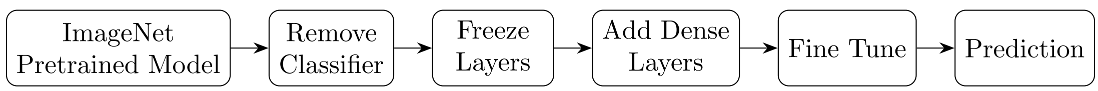

Figure 2: Transfer Learning Workflow

Advantages

- Faster convergence.
- Higher accuracy on small datasets.
- Reduced training time.
- Lower computational cost.

14 Dataset

Dataset: CIFAR-10

- Training Images: 50,000
- Testing Images: 10,000
- Number of Classes: 10
- Image Size: $32 \times 32 \times 3$
- Classes: Airplane, Automobile, Bird, Cat, Deer, Dog, Frog, Horse, Ship and Truck.

15 Experimental Procedure

Task 1: Dataset Preparation

1. Load the CIFAR-10 dataset using TensorFlow/Keras.
2. Normalize the pixel values to the range [0,1].
3. Display ten sample images.
4. Print the dimensions of the training and testing datasets.

Task 2: Transfer Learning
Load any one of the following pretrained CNN models.

- VGG16
- ResNet 50
- MobileNet V2
- Efficient Net B0 (Optional)

Perform the following steps.

1. Load pretrained ImageNet weights.
2. Remove the original classification layer.
3. Freeze the convolutional base.
4. Add a Global Average Pooling layer.
5. Add a Dense layer with ReLU activation.
6. Add the output layer with Softmax activation.

Task 3: Model Training
Train the model using the following training parameters:

Parameter	**Value**
Optimizer	Adam
Learning Rate	0.001
Batch Size	32
Epochs	10–20
Loss Function	Categorical Cross Entropy
Evaluation Metric	Accuracy

Task 4: Fine Tuning

1. Unfreeze the last convolution block.
2. Train for another 5-10 epochs.
3. Compare the classification accuracy before and after fine tuning.

Task 5: Model Evaluation
Evaluate the trained model using

- Accuracy
- Precision
- Recall
- F1-score
- Confusion Matrix
- Classification Report

16 Hyperparameter Study

Compare the performance using different settings.

Hyperparameter	**Values**
Learning Rate	0.001, 0.0001
Batch Size	16, 32, 64
Epochs	10, 20
Optimizer	Adam, SGD
Dense Units	128, 256
Frozen Layers	All, Partial

17 Source Code

The complete source code, implementation details, and execution notebooks for this experiment are available at the following repository:

`https://github.com/KalpanaBhaskar/Deep-Learning-Laboratory/tree/main/4_DLM`

18 Mandatory Plots

1. Sample CIFAR-10 Images

Figure 3: Sample CIFAR-10 images across different classes.

Interpretation: The dataset contains 10 visually distinct classes with 32×32 RGB images. Even at low resolution, class-specific features like shape and colour are visible. This confirms the dataset is suitable for evaluating CNN architectures across varied object categories.

2. Training and Validation Accuracy

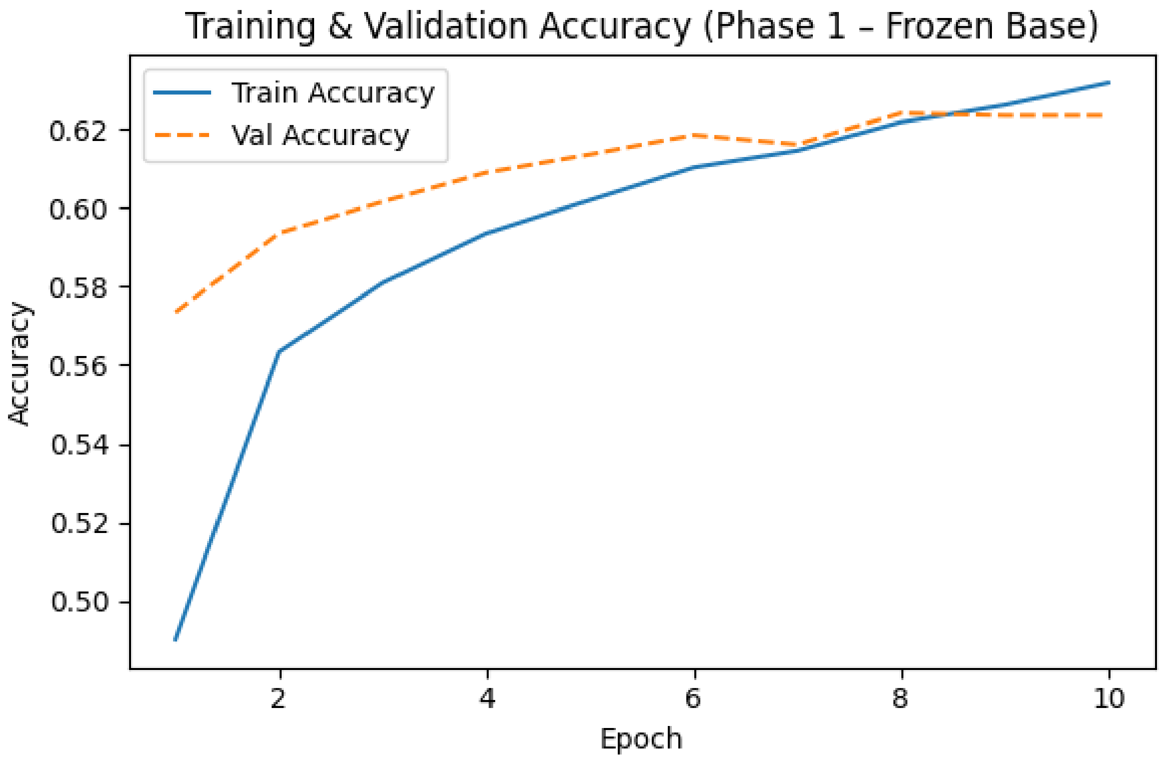

Figure 4: Training Accuracy vs. Epochs.

Interpretation: Both training and validation accuracy rise steadily from 0.50 to 0.63 over 10 epochs without any divergence between the two curves. This indicates the frozen pretrained backbone is extracting useful features and the new classifier head is learning well without overfitting, which is the expected behaviour in feature extraction mode.

3. Training and Validation Loss

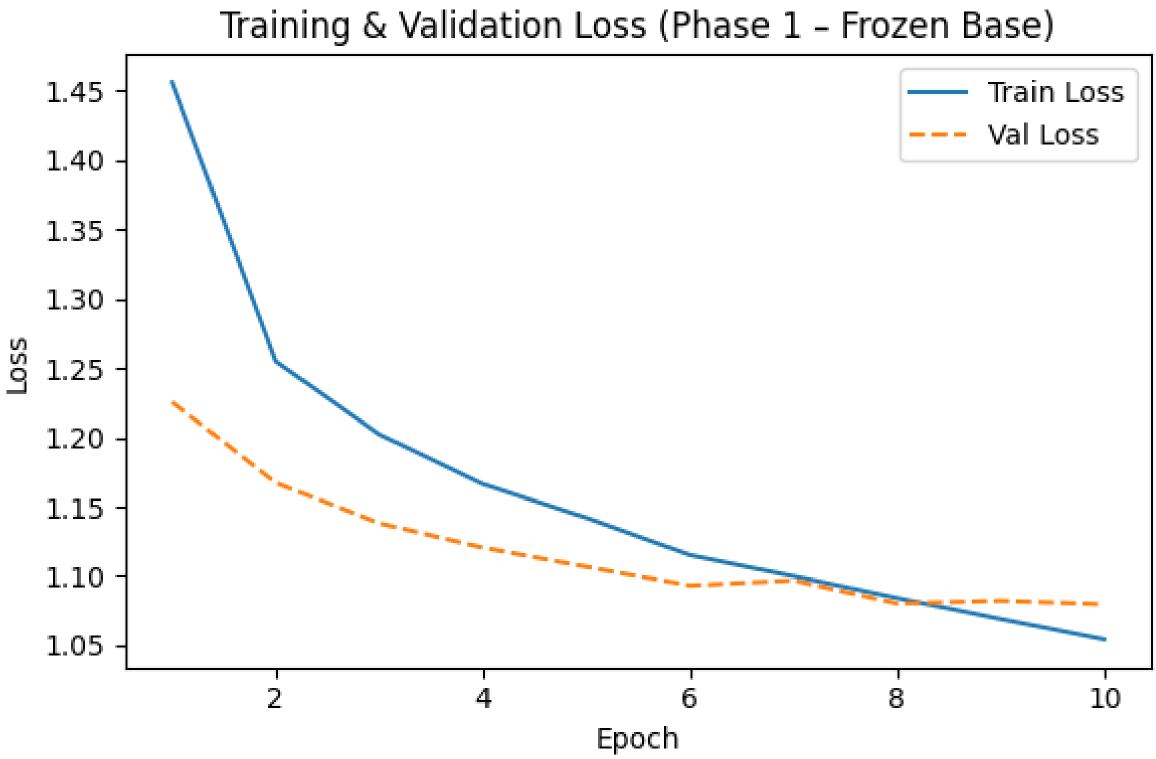

Figure 5: Training Loss vs. Epochs.

Interpretation: Training loss drops sharply from 1.45 to 1.05 while validation loss follows closely, settling around 1.08. The small gap between the two curves confirms the model is generalising well and not memorising training data, a sign that freezing the convolutional base effectively acts as a regulariser.

4. Confusion Matrix

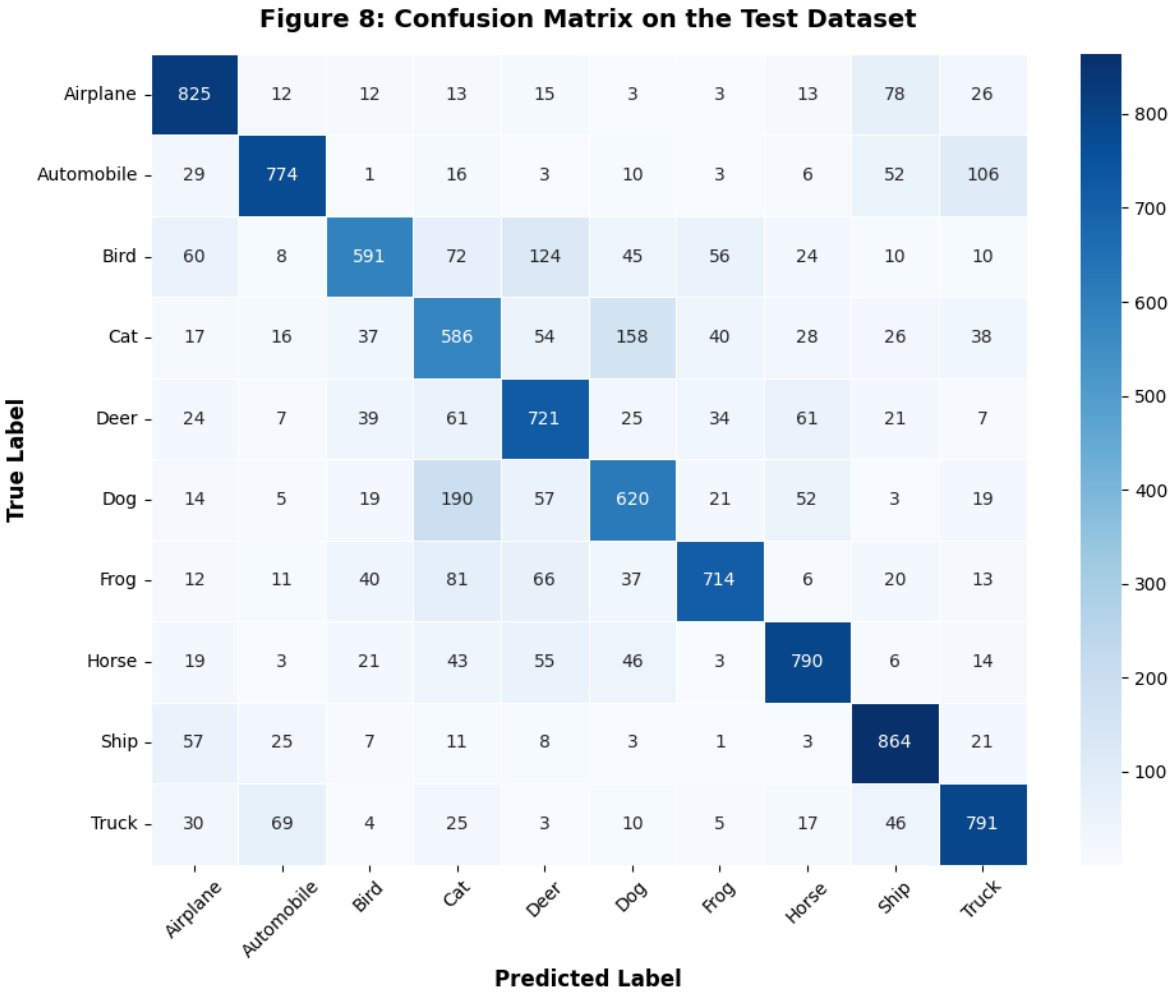

Figure 6: Confusion Matrix on the test dataset.

Interpretation: Ship (864/1000) and Automobile (774/1000) are the best-classified classes due to their distinct shapes and colour profiles. Cat and Dog show the most confusion with each other (158 Cat→Dog, 190 Dog→Cat), which is expected since both share similar fur textures, body proportions and poses at 32×32 resolution. Overall diagonal dominance confirms the model has learned meaningful class-level representations.

5. Misclassified Images

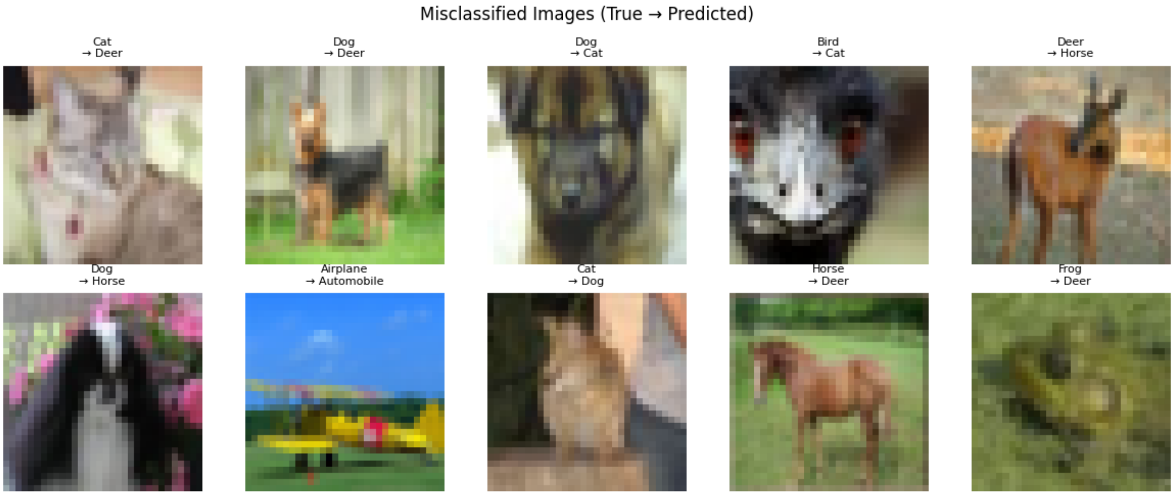

Figure 7: Examples of misclassified test images.

Interpretation: Most misclassifications occur between visually similar classes — CatDog, DeerHorse, and BirdCat — where shared low-level features like fur texture and body outline mislead the classifier. The 32×32 resolution makes fine-grained distinctions harder since breed-specific details are lost during downsampling. This suggests the model relies on texture and colour rather than structural features, which is a known limitation of transfer learning on low-resolution data.

19 Results

19.1 Performance Metrics

Metric	Value
Training Accuracy	83.37%
Testing Accuracy	76.39%
Precision	0.7639
Recall	0.7639
F1-score	0.7633
Training Time	284.2 seconds
Total Parameters	14,848,586

19.2 Comparison of CNN Architectures

Model	Parameters	Accuracy (%)	Training Time
LeNet-5	60K	33.00%	63.0 s
AlexNet	61M	54.77%	163.2 s
VGG16	138M	62.14%	191.7 s
GoogleNet	6.8M	68.92%	340.5 s
ResNet50	25.6M	76.39%	262.2 s

20 Discussion Questions

1. **Why is AlexNet considered a breakthrough in deep learning?**

AlexNet demonstrated that deep CNNs could achieve high accuracy on large-scale image classification using GPUs. It introduced practical use of ReLU and dropout, enabling faster training and reduced overfitting. Its success in the 2012 ImageNet competition established deep learning as a powerful approach for computer vision.

2. **Why does VGG16 use only 3×3 convolution filters?**

   VGG16 uses 3×3 filters to capture detailed spatial features with fewer parameters. Multiple 3×3 convolutions can provide a larger effective receptive field; for example, two 3×3 layers approximate a 5×5 receptive field while adding more nonlinearities. This increases network depth and representation power efficiently.

3. **Explain the advantages of the Inception module.**

   The Inception module applies different operations such as 1×1, 3×3, and 5×5 convolutions and pooling in parallel. This captures features at multiple spatial scales. 1×1 convolutions also reduce the number of channels, lowering computational cost while producing richer feature representations.

4. **What is the purpose of residual learning?**

   Residual learning makes very deep networks easier to train by addressing the degradation problem. ResNet uses skip connections that add the input directly to the layer output. This improves gradient flow during backpropagation and enables effective training of much deeper networks.

5. **Differentiate LeNet and ResNet.**

   LeNet is an early, shallow CNN designed mainly for handwritten digit recognition, with few layers and parameters. ResNet is a much deeper CNN designed for large-scale image recognition and uses residual/skip connections. Thus, LeNet is lightweight and suitable for simple tasks, while ResNet provides greater representation capacity for complex vision tasks.

6. **What is Transfer Learning?**

   Transfer learning reuses knowledge learned from a pretrained model for a related task. A model trained on a large dataset such as ImageNet provides useful features such as edges, textures, and shapes. This reduces training time and improves performance, especially when the new dataset is small.

7. **Why is fine tuning required?**

   Fine tuning adapts pretrained features to the new dataset or task. While early layers learn general features, deeper layers learn task-specific representations. Updating selected pretrained layers allows the model to learn features better suited to the target task while requiring less training than training from scratch.

8. **Explain the difference between dilated convolution and transpose convolution.**

   Dilated convolution expands the receptive field by inserting gaps between filter elements without significantly increasing parameters or reducing feature-map resolution. It is commonly used in semantic segmentation. Transpose convolution increases spatial resolution by learning an upsampling operation and is commonly used in encoder–decoder architectures and image generation.

9. **Why do pretrained models converge faster?**

   Pretrained models start with weights that already encode useful features learned from large datasets. Unlike randomly initialized models, they do not need to learn basic features such as edges and textures from scratch. This provides a better optimization starting point, reducing training iterations and often improving performance with limited data.

10. **Compare the computational complexity of LeNet and ResNet.**

    LeNet has much lower computational complexity because it contains fewer layers, parameters, and smaller feature maps. ResNet is considerably more computationally expensive due to its deeper architecture and larger feature representations. However, residual connections enable effective training of deep networks. Thus, LeNet is lightweight, while ResNet provides higher representation capacity at greater computational cost.

21 Additional Exercises

1. Implement transfer learning using VGG16.
2. Repeat the experiment using ResNet50.
3. Compare Adam and SGD optimizers.
4. Compare training with frozen layers and fine tuning.
5. Evaluate the model using Precision, Recall and F1-score.
6. Compare the performance of LeNet, AlexNet and ResNet.

22 References

1. Yann LeCun, Leon Bottou, Yoshua Bengio and Patrick Haffner, Gradient-Based Learning Applied to Document Recognition, Proceedings of the IEEE, 1998.
2. Alex Krizhevsky, Ilya Sutskever and Geoffrey Hinton, ImageNet Classification with Deep Convolutional Neural Networks, NeurIPS, 2012.
3. Karen Simonyan and Andrew Zisserman, Very Deep Convolutional Networks for Large-Scale Image Recognition, ICLR, 2015.
4. Christian Szegedy et al., Going Deeper with Convolutions, CVPR, 2015.
5. Kaiming He, Xiangyu Zhang, Shaoqing Ren and Jian Sun, Deep Residual Learning for Image Recognition, CVPR, 2016.
6. Ian Goodfellow, Yoshua Bengio and Aaron Courville, Deep Learning, MIT Press, 2016.
7. TensorFlow Documentation: `https://www.tensorflow.org`
8. Keras Documentation: `https://keras.io`